Soccer Sage

Forecasting the Future of Soccer

Team P28
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ALDA FALL 24 – NCSU

Background & Motivation

- Modern soccer generates massive data
  - 3M data points per match (StatsPerform)
  - 75+ metrics per player (StatsPerform)
  - Over €10B transfer market (2023) (TransferMarkt)

- **Success story**
  - Liverpool FC used data-driven approach for transfers (Medium)

References

- [1] Ulmer, Ben, Matthew Fernandez, and Michael Peterson. "Predicting soccer match results in the English premier league." Doctoral dissertation, Doctoral dissertation, Ph. D. dissertation, Stanford (2013).
  - Accuracy: 50%
- [2] L. Hervert-Escobar, T. I. Matis and N. Hernandez-Gress, "Prediction Learning Model for Soccer Matches Outcomes," *2018 Seventeenth Mexican International Conference on Artificial Intelligence (MICAI)*, Guadalajara, Mexico, 2018.
  - Accuracy: 47.92%
- [3] Holmes, Benjamin, and Ian G. McHale. "Forecasting football match results using a player rating based model." *International Journal of Forecasting* 40.1 (2024): 302-312.
  - Accuracy of best model: 53.88%
- [4] Shin, J., & Gasparyan, R. (2014). A novel way to soccer match prediction. Stanford University: Department of Computer Science.
- [5] Luca Pappalardo, Paolo Cintia, Paolo Ferragina, Emanuele Massucco, Dino Pedreschi, and Fosca Gian-notti. 2019. PlayeRank: Data-driven Performance Evaluation and Player Ranking in Soccer via a Machine Learning Approach. ACM Trans. Intell. Syst. Technol. 10, 5, Article 59 (September 2019).

Dataset Overview

- **Dataset**
  - European Soccer Database (2008-2016) comprising of 5 major leagues, 10,000+ matches, 25,000+ players.
  - The dataset's rich, interconnected information on matches, player attributes, and team metrics allows us to extract valuable insights.

- **Can we use the dataset to**
  - Predict match outcomes?
  - Identify optimal player transfers?
  - Optimize team's performance?

Our Approach

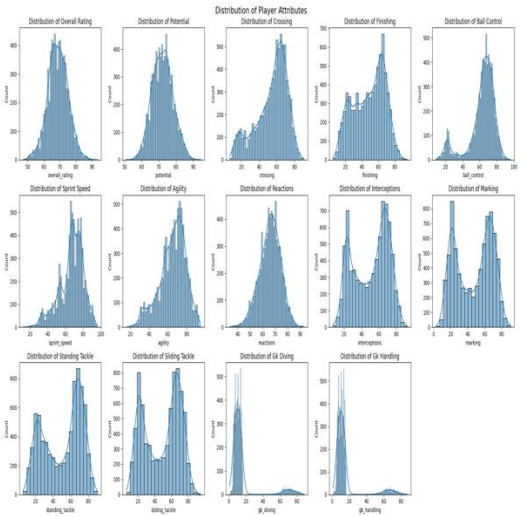
- **Methodology**
  - Comprehensive Exploratory Data Analysis (EDA)
  - Data Preprocessing
  - Data Transformation
  - Prediction of objectives
- **Prediction of Match Outcomes**
  - Neural Networks
  - Random Forest Classification
  - Gradient Boosting
- **Prediction of Player Transfers**
  - Filter players that have high rating and high potential
  - Forecast player ratings to show future potential and usefulness for the team

Data Preprocessing & Transformation

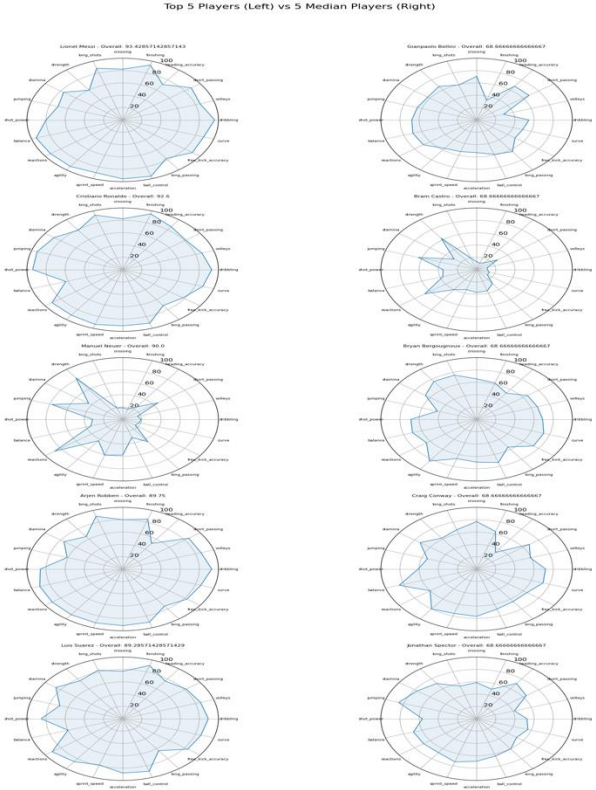
- **Steps**
  - **Merging** tables using unique identifiers.
  - Handling **missing values** with average fills.
  - **Aggregating** player, team data to analyze performance over multiple years.

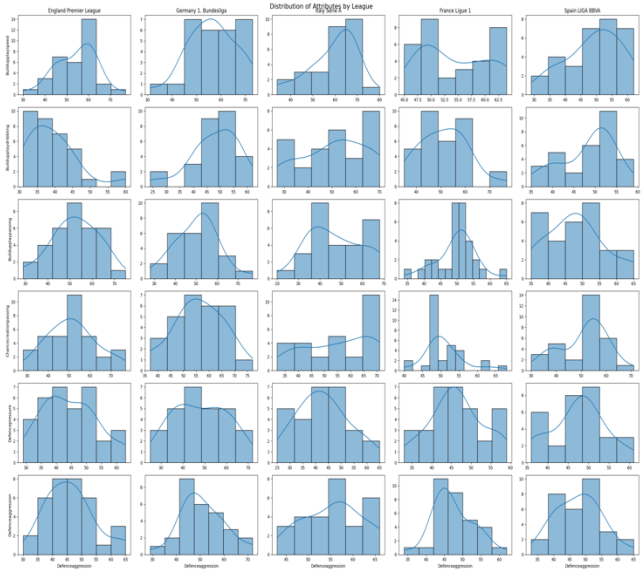
- **Feature Engineering**
  - Home/Away team metrics based on players ratings.
  - Match outcome encoding: Home Win (1), Draw (0), Loss (-1).
  - Tactical attributes (Numerical and Categorical) encoded for model training.

EDA Insights



Skill metrics (e.g., passing, shooting) exhibit a bell curve.

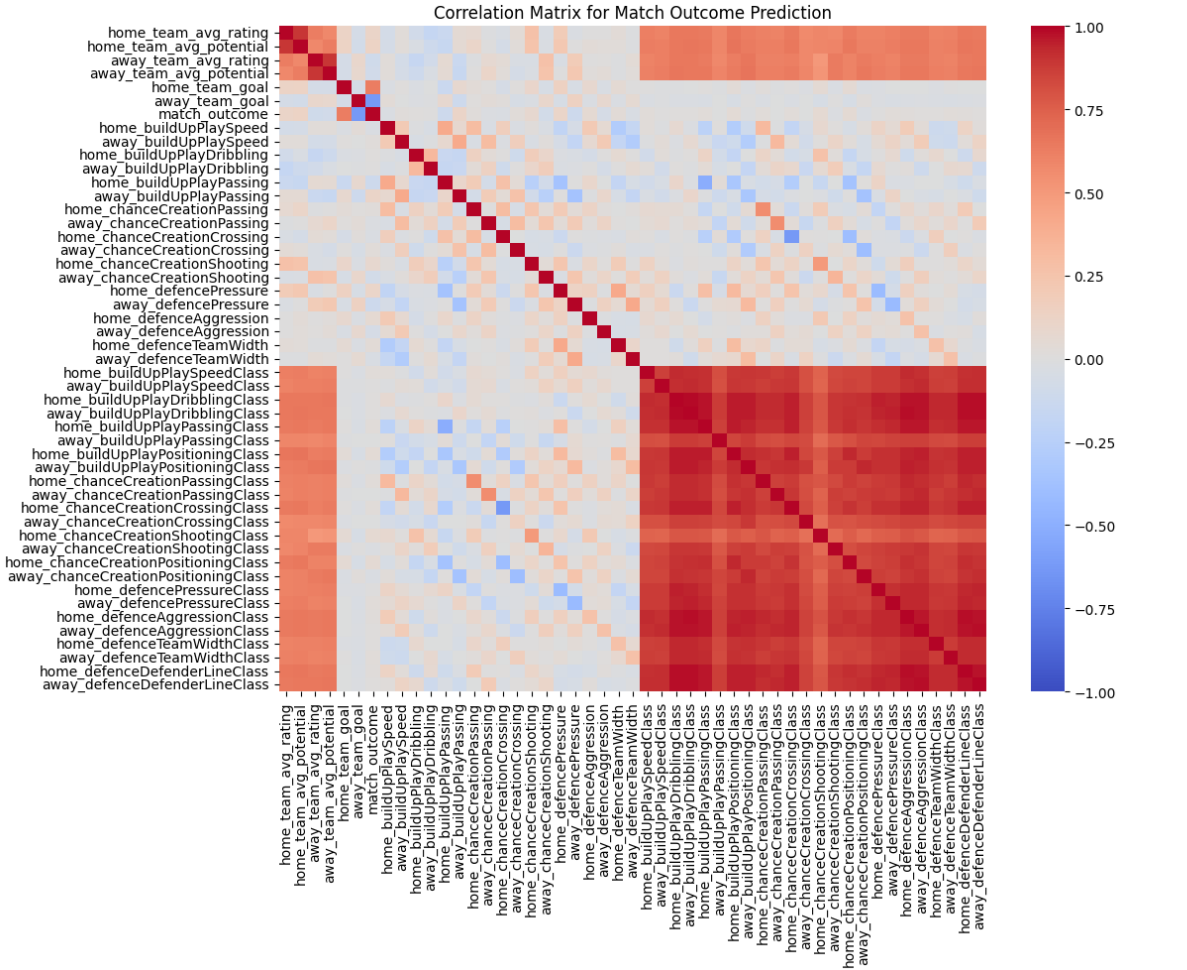




- **League characteristics:**
  - La Liga emphasizes technical passing.
  - Bundesliga: Fast-paced build-up.
  - Ligue 1: Defensive intensity.

EDA Insights

We can see that lots of features are highly correlated



Prediction Strategy for Match Outcome

1. **Random Forest Classifier**
   - **Purpose**: Aggregates multiple decision trees to reduce overfitting.
   - **Parameters**: n_estimators=100, class_weight='balanced'
   - **Advantages**: Effective for multiclass classification, Reduces overfitting, Implicit feature selection

2. **Gradient Boosting Classifier**
   - **Purpose**: Trains sequential models to minimize classification errors by combining weak learners.
   - **Parameters**: n_estimators=100
   - **Advantages**: High Accuracy, Suitable for complex data patterns

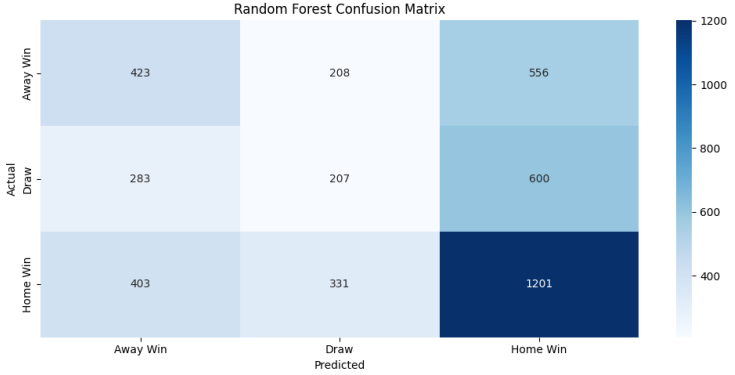
3. **Neural Network with CrossEntropy Loss**
   - **Architecture**: 2 layers (500, 100 neurons)
   - **Parameters**: hidden_layer_sizes=(500, 100), max_iter=500
   - **Loss**: Standard CrossEntropy Loss

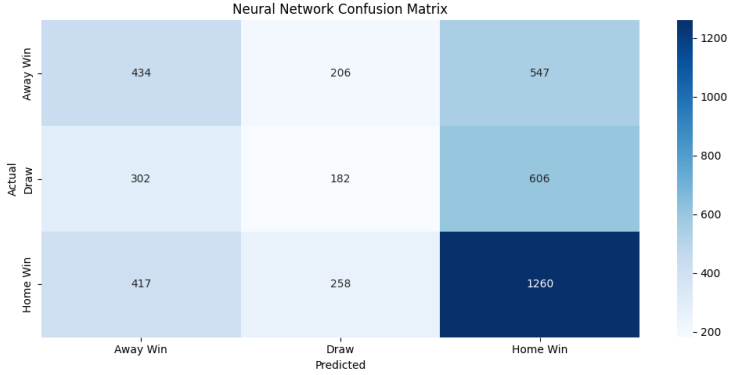
4. **Neural Network(Pytorch) with Focal Loss:**
   - **Architecture**: 5 layers (128, 256, 128, 64 neurons) with dropout
   - **Parameters**: Epochs=50, Batch_size=32, Learning_rate=0.001, Weight_decay
   - **Advantages** : Focuses on hard-to-classify, underrepresented samples

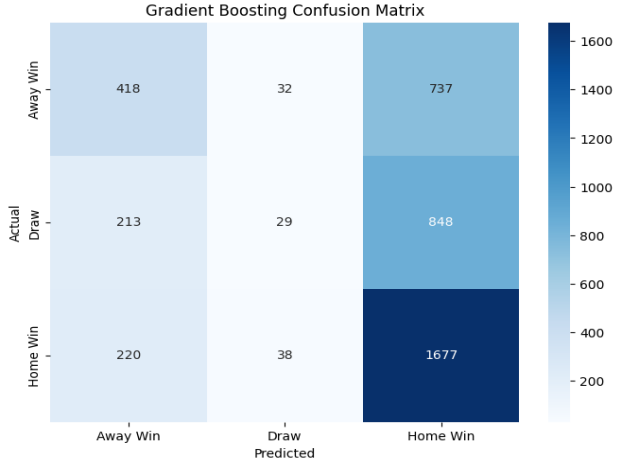
Prediction Strategy for Player Transfers

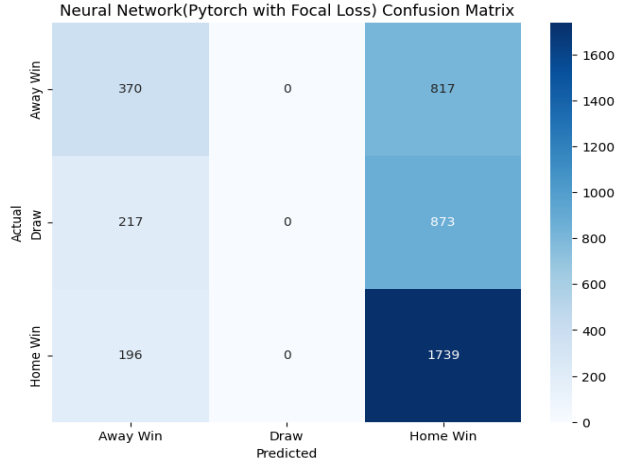
1. **Data-Driven Team Weakness Identification**
   - Analyzes goals scored/conceded and pre-2016 team stats (e.g., build-up, chance creation).
   - Defines **target roles** (attacker, defender, midfielder, goalkeeper) based on **deficiencies**.
2. **Targeted Player Filtering and Ranking**
   - Filters players by **role-specific metrics** using computed mean thresholds.
   - Identifies **top-performing** and **high-potential** candidates based on dynamic thresholds.
3. **Predictive Modeling for Future Potential**
   - Uses **linear regression** on historical data to project future player potential.
   - Applies latest stats to generate projected potential, highlighting long-term growth.

Results for Match Outcome Prediction









Results for Match Outcome Prediction

Table 1: Comparison of Model Results

Metric	**Random Forest**		**Gradient Boosting**		**Neural Network**		**NN + Focal Loss**	
	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
Overall Accuracy	0.4482	0.43	0.5043	0.42	0.4454	0.43	0.5007	0.40
Class	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
Away Win	0.41	0.38	0.49	0.35	0.38	0.37	0.47	0.31
Draw	0.26	0.17	0.29	0.03	0.28	0.17	0.00	0.00
Home Win	0.53	0.65	0.51	0.87	0.52	0.65	0.51	0.90
Macro Avg	0.40	0.39	0.43	0.37	0.39	0.39	0.33	0.34
Weighted Avg	0.42	0.43	0.45	0.42	0.42	0.43	0.37	0.40

Results

Club: Liverpool FC
Weakness: Attack

Role	Player Name	Overall Rating	Potential	Finishing	Interceptions	Short Passing
Top Performers - Attackers	Lionel Messi	94	96	94	22	89
	Cristiano Ronaldo	92.6	92.6	95	27	81.4
	Wayne Rooney	91.5	92	91.5	78	85
	Thierry Henry	91	92	95	83	84.5
	Ronaldinho	91	94	90	81	90
High Potential - Attackers	Toni Kroos	75	90	71	68	84
	Juan Carlos Valeron	78	88.5	70	57.5	85.5
Top Performers - Defenders	Alessandro Nesta	91	92	24	92	54
	John Terry	90.5	90.5	26	86	58.5
	Nemanja Vidic	89	89	37	83	68.5
High Potential - Defenders	Douglas Costa	75	90	68	43	78
	Youri Tielemans	74.3	88.8	59.9	68.3	76.7
Top Performers - Midfielders	Lionel Messi	94	96	94	22	89
	Cristiano Ronaldo	92.6	92.6	95	27	81.4
High Potential - Midfielders	Jose Baxter	71	90	68.5	65	67
	Nikola Zigic	79.5	90	93	70	81
Top Performers - Goalkeepers	Gianluigi Buffon	92	93	15.5	90	23
	Iker Casillas	91	92	22	87	21
High Potential - Goalkeepers	Anton Mitryushkin	64	80	17	16	27
	Luca Lezzerini	62	80	25	25	31

Conclusion

- Random Forest has the lowest accuracy, but it is a very simple model to implement and train, and it achieves a good F1-Score.
- Gradient Boosting provides one of the best accuracies while maintaining a high F1- score, showcasing the power of ensemble models.
- Neural networks tend to overfit the data and increasing the hidden layer size leads to a dramatic decrease in the F1-score due to the underrepresentation of the ‘Draw’ class.
- Player transfer recommendations are an initial step for scouting reports.

Any **Scope for improvement**?

- Handle class imbalance while maintaining accuracy.
- Try ensembling different classifiers using voting methods to improve accuracy.
- Improve player transfer recommendations by incorporating more features.

Questions?

Thank You!